



SCIENTIFIC INSTRUMENTS

SI-7000

Tank Gauging System

DCS Modbus

Interface Specification

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1 General Description

This document defines a limited implementation of the Modbus protocol for the Scientific Instruments SI-7000 LTD. This document should provide programmers with the details necessary to prepare and test the user interface on a host computer.

The LTD is a tank gauging system designed to measure level, temperature and density in large tanks of cryogenic liquids (e.g. liquid natural gas). A probe is driven up and down in a tank to collect temperature and density data, as specified by the operator. The data is stored in memory for output to the host computer on demand. The programmer should read and be familiar with the Operation Manual for the LTD, as the basic system operation information is not duplicated here.

In the general Modbus specification, certain characteristics are fixed, such as frame format and frame sequence. Other characteristics are variable, including the type of interface (e.g. RS232, RS485), baud rate, parity, number of stop bits and transmission mode (ASCII or RTU). In the LTD implementation of the Modbus interface, the interface type is RS485. (To interface with the host computer using RS232, a converter must be used.) The baud rate is fixed at 9600 baud, and the other parameters are odd parity, 8 data bits, and 1 stop bit. The transmission mode used is RTU.

The Modbus interface allows the host computer access to the LTD's current mode of operation, the direction and speed of the probe, individual system status and alarm bits, profile data, and current tank conditions. The mode of operation can be controlled via Modbus, and the probe can be driven manually at several different speeds.

Output of level, temperature, and density will be scalable in future firmware versions.

In the last section, programming suggestions are given to assist the programmer in handling important information and status bits appropriately.

Note:

The minimum time for the Command/Request from Host (PC) to Device (LTD) should be 1 Second.

2 Data Addressing & Data Definition

A number of Modbus functions are utilized in the LTD Modbus interface. Function 1 is used to read system mode, motor drive direction, and speed, while Function 5 is used to set them. Function 2 is used to read individual status and alarm bits. Function 3 is used to read profile and alarm configuration registers and Function 6 makes it possible to set them. Function 4 allows the host computer to read sensor data from the LTD.

When more than one tank is connected to a host computer, each tank must be addressed and scanned individually. The Device ID is the Tank ID of each system, as explained below in the Communication Settings section of the Hardware Specifications.

2.1 Mode, Speed (Digital Output or Input, 1 Bit) (Functions 1, 5)

Mode, Speed Base Address = 0001

Maximum Legal Address = 0016

The bits are defined as follows:

Parameter	Address
Manual	00001
Calibrate	00002
Auto	00003
Profile	00004
Reserved	00005
Reserved	00006
Reserved	00007
Reserved	00008
Stop	00009
Up Slow	00010
Up Medium	00011
Up Fast	00012
Down Slow	00013
Down Medium	00014
Down Fast	00015
Reserved	00016

The information shown above gives the current LTD mode of operation and the probe's movement and can be accessed using Function 1. These are individual bits, not registers. Any of these quantities may be set using Function 5 with the same address.

Bits 1 – 5 indicate the current mode of operation. Each mode is mutually exclusive, so that if one bit is set, the others are cleared. When using Function 5 to change the system mode, it is only necessary to set the bit corresponding to the new mode. It is not necessary (or possible) to manually clear the bit corresponding to the old mode.

Bits 9 – 15 indicate the probe's direction and speed of travel. Again, each drive speed is mutually exclusive, so if the host computer sets one bit using Function 5, the others are automatically cleared.

These bits may be read to determine current system status at any time without any effect on the system mode of operation, but it should be noted that manually setting any of the bits relating to the probe's direction and speed of travel will halt any profile activity or liquid level tracking that is taking place and put the system in the Manual drive mode. The system will continue indefinitely in this mode until it is set back to "Auto", "Cal", or "Profile." The system will not be able to perform any of its automatic functions while it remains in the Manual mode. One suggestion to bring attention to this condition would be to indicate the manual mode in red to alert the operator that this is not the normal mode of operation. The "Auto" mode, on the other hand, could be indicated in green, since this is the normal mode of operation where liquid level will be tracked continuously.

2.2 Status, Alarms (Digital Input, 1 Bit) (Function 2)

The main purpose of this document is to provide the information necessary for a programmer to develop a user interface for the LTD on a host computer.

Status Base Address = 10001

Alarms Base Address = 10017

Maximum Legal Address = 10032

2.2.1 Status Bits

The status bits provide important information about system operation and are essential for verifying that data reported by the LTD is current and valid. Each status bit (or status indicator) is given along with its address. They are defined as follows:

Status Indicator	Address	
Bottom Reference	10001	
Lower Level Sensor	10002	
Upper Level Sensor	10003	
Interlock	10004	
Profile Complete	10005	
Metric	10006	
Reserved	10007	
Reel Alarm Disable	10008	
Reel Alarm	10009	
Probe Un-calibrated	10010	
Reserved	10011	
Interval Timer	10012	
Probe At Liquid Level	10013	
Reserved	10014	
Reserved	10015	
Reserved	10016	

Some of these status indicators provide information that is more important than others for plant operators to see. The ones that are essential and should definitely be reported in any custom user interface are Bottom Reference, Lower Level Sensor, Upper Level Sensor, Interlock, Reel Alarm, Probe Un-calibrated, Interval Timer, and Probe At Liquid Level. The others are optional, as explained below.

Bottom Reference is activated whenever the bottom reference switch is closed, indicating that the probe is on the bottom of the tank. **Lower Level Sensor** and **Upper Level Sensor** are set whenever the sensors are in liquid. These are important for observing the physical conditions around the probe and verifying where it is in the tank.

Interlock is set whenever the probe is at the maximum allowed probe position; the probe will not drive up any more when it reaches Interlock. It usually indicates some kind of error condition if the probe reaches Interlock during normal operation, since the probe should be at liquid level or below. This could also happen if an operator drives the probe up manually.

When **Reel Alarm** is set, it shows that a physical drive problem has occurred. This indicates the need for investigation by maintenance personnel and is absolutely essential for monitoring proper operation of the equipment.

Probe Un-calibrated is activated whenever it is possible that the current probe position is not correct. This status indicator is also important, since it means that the liquid level value reported may be incorrect as well.

When **Interval Timer** is set, it indicates that the system will perform profiles automatically at the programmed interval. If not set, profiles can be run manually.

Probe At Liquid Level is set whenever the system is in Auto and the probe is at the liquid/vapor interface. This means that the level reported is current.

The status indicator, **Profile Complete**, may be of more importance to the programmer than to the operator. When this indicator is cleared, the LTD is in the process of running a profile. When this indicator is set, it means that the profile is complete and the data can be collected and analyzed. The remaining status indicators are not normally needed by operators, but they are useful for someone checking the complete status of the equipment. **Reel Alarm Disable** should not normally be set, because this overrides a safety feature in the equipment that could prevent possible damage in the event of equipment failure. Having an alarm tied to this indicator would help prevent it from being set without operator knowledge. The status indicator, **Metric**, will be set if values are reported in metric units, and cleared if values are reported in English units. Once this is set, it should not need to be changed.

2.2.2 Alarms

The alarm bits relate to conditions of liquid in the tank and are important with respect to plant operation. Each alarm is given along with its address. The alarms are defined as follows:

Alarm	Address	
Low Density	10017	
High Density	10018	
Low Temperature	10019	
High Temperature	10020	
Low, Low Level	10021	
High, High Level	10022	
Low Level	10023	
High Level	10024	
Profile Temperature Deviation	10025	
Profile Density Deviation	10026	
Reserved	10027	
Reserved	10028	
Profile Low Temperature	10029	
Profile High Temperature	10030	
Profile Low Density	10031	
Profile High Density	10032	

2.3 Profile & Alarm Control (Analog Output or Input) (Functions 3, 6)

Functions 3 and 6 are used to read and control profile settings and product alarm settings. This makes it possible for operators to completely control the profiles and change alarm settings from the DCS.

2.3.1 Addressing

Base Address = 40001

Maximum Address = 40023

Parameter	Address
Profile First Point	40001
Profile Increment	40002
Profile Dwell Time	40003
Reserved	40004
Reserved	40005
Reserved	40006
Reserved	40007
Reserved	40008
Reserved	40009
Automatic Profile Interval	40010
Automatic Profile Enable	40011
Automatic Profile Hour	40012
Automatic Profile Minute	40013
Low Density Alarm Set Point	40014
High Density Alarm Set Point	40015
Low Temperature Alarm Set Point	40016
High Temperature Alarm Set Point	40017
Low Low Level Alarm Set Point	40018
High High Level Alarm Set Point	40019
Low Level Alarm Set Point	40020
High Level Alarm Set Point	40021
Temperature Deviation Alarm Set Point	40022
Density Deviation Alarm Set Point	40023

2.3.2 Definition

When a profile is initiated, the probe drives to the tank bottom and starts the profile there. **Profile First Point** is the first programmed point at which the probe will stop on the way up to take temperature and density measurements. (Actually it is the second point in the profile, since the first point is always at the bottom of the tank.)

Profile Increment is the distance that is added to the first point and each successive point to determine where to stop and take readings on the way back to the surface of the liquid.

Profile Dwell Time is the time in seconds that the probe will pause at each point to allow readings to stabilize before recording the value.

A profile can be started manually by switching the mode to Profile Mode. Alternatively, a profile can be run automatically at either a fixed time each day or at regularly repeated intervals.

Automatic Profile Interval designates the number of minutes between each successive profile run (after the first profile starting at **Automatic Profile Hour** and **Automatic Profile Minute**).

Automatic Profile Enable designates whether profiles will be run automatically or not. A value of 0 disables automatic profiles. A value of 1 causes profiles to be initiated at the programmed time each day and at the interval chosen. If only one profile is desired per day, the interval should be 1440 minutes (to start it the next day at the same time). The first profile (after enabling the feature) will be at the time defined in **Automatic Profile Hour** and **Automatic Profile Minute**.

The remaining quantities set the alarm limits for low and high density values, temperature values, liquid level, and temperature and density deviations. **Temperature Deviation Limit** is the amount of change in temperature from one point to the next in a profile that will cause an alarm. **Density Deviation Limit** is the amount of change in density from one point to the next in a profile that will cause an alarm.

2.3.3 Scaling

In the LTD implementation of the Modbus specification for Functions 3, 4, and 6 the values are communicated to the DCS using 16 bit signed or unsigned integers, so it must be decided what each integer will represent. Each quantity is scaled between high and low limits for the value being reported, along with high and low scale values. These limits are currently fixed. (In a future firmware version it will be possible to change them, and the proper values for each site will be established during commissioning if not previously set.)

Separate scaling factors are set for level, temperature, and density. The following table defines the scaling.

	Low Limit	High Limit	Low Scale	High Scale	Value to be reported from LTD	Value Sent to DCS
Level (m)	0 m	65.000 m	0	65000	15.000 m	15000
Temperature	-327.68	327.67 °C	-32768	32767	-160.00	-16000
Density (kg/m ³)	0	655.35 kg/m ³	0	65535	450.00	45000
Level (ft)	0 ft	200.00 ft	0	60000	100.00 ft	30000
Temperature	-327.68	327.67 °F	-32768	32767	-256.00 °F	-25600
Density (lb/ft ³)	0	65.535 lb/ft ³	0	65535	27.500	27500

For Level (in metric units), the 16-bit integer value in the Modbus register would read directly in millimeters of liquid level as an unsigned integer. Each count of the Modbus integer would represent 1 mm. For temperature (both English and metric), each count would represent 100th's of a degree in each respective scale, and would be reported as a signed integer.

When reading or writing quantities relating to level (such as Profile First Point, Profile Increment, and High Level Alarm Set Point), use the scaling factors for Level. This applies both to quantities received and transmitted. Quantities received from the host computer are interpreted according to the current scaling values in effect. Dwell times are in seconds.

2.4 Sensor Data, Profile Data (Analog Input) (Function 4)

The current conditions in the tank at the location of the probe and information about the condition of the tank during the last profile are given by the sensor data and profile data quantities.

2.4.1 Addressing

Base Address for Sensor Data = 30001

Maximum Legal Address = 30620

Parameter	Address
Current Probe Position	30001
Current Temperature	30002
Current Density	30003
Liquid Level	30004
Reserved	30005
Number of Points	30006
Profile Month	30007
Profile Day	30008
Profile Hour	30009
Profile Minute	30010
Current Hour	30011
Current Minute	30012
Current Second	30013
Reserved	30014
Reserved	30015
Reserved	30016
Reserved	30017
Reserved	30018
Reserved	30019
Reserved	30020
Profile Probe Position Point 0	30021
Profile Temperature Point 0	30022
Profile Density Point 0	30023
Profile Probe Position Point 1	30024
Profile Temperature Point 1	30025
Profile Density Point 1	30026
Profile Probe Position Point n	$30021 + (3 * n)$
Profile Temperature Point n	$30022 + (3 * n)$
Profile Density Point n	$30023 + (3 * n)$
Profile Probe Position Point 199	30618
Profile Temperature Point 199	30619
Profile Density Point 199	30620

Note: The term “profile probe position” is used instead of “profile level,” because in this document, level is usually used to indicate the actual liquid level. Only the last point in a regular profile is actually at “liquid level.”

2.4.2 Definition

Current Position, Current Temperature, and Current Density are the current values where the probe is now, whether at the bottom of the tank or at liquid level. **Liquid Level** holds the most recent liquid level reading. If the probe is beneath the surface, such as during a profile, the liquid level reading will not be updated until the probe returns to the surface. The status indicator “At Liquid Level” (described in a previous section) is used to show when the value reported is up to date.

Number of Points contains the number of profile points collected during the last profile. The number of points is determined by the location of the programmed starting point, the increment added, and liquid level. The first point in a profile is always the bottom of the tank. The next point is programmable, and is usually set to an whole number such as 1 m, and then a programmable increment is added (also normally 1 m). This puts profile points at whole numbers, such as 1 m, 2 m, 3 m, etc. The probe moves up, collecting data until liquid level is reached (or the maximum number of points is reached, causing the profile to terminate prematurely). The last point is normally at liquid level.

The maximum number of points allowed is 500 (0 to 499). To limit the number of points to a lower value, set the profile increment so that at maximum liquid level, the number of points will not exceed the desired limit. For example, if maximum liquid level is 30 m and the profile increment is 1 m, the maximum number of points would be 31 (including the bottom reference point) and the total number of registers needed for position, temperature and density would be 93 (31 x 3).

Profile Month, Profile Day, Profile Hour, and Profile Minute show the date and time when the first profile point is collected. **Current Hour, Current Minute, and Current Second** show the current system time. One way to constantly verify communications with the LTD is to monitor the system time and verify that it is always changing. If it stops changing, an alarm should be generated.

The probe position, temperature, and density for each point collected in a profile are available at the addresses shown. Data in registers beyond the number of points collected will be zero.

2.4.3 Scaling

The scaling for Function 4 values is controlled by the same parameters that are used for values in Functions 3 & 6. Please refer to that section for more information about scaling.

3 Hardware Specifications

3.1 General

A host computer may be connected directly to the LTD using the RS485 field wiring. A site-wiring diagram is generally provided for each site. In order to establish communications, there are a number of data parameters that must be set properly.

3.2 Communication Settings

In one possible configuration, the LTD is connected to the Tank Gauge Interface Module (TGIM). The TGIM serves as a power switch and power indicator for the gauge and a connection point for all field wiring. It is usually located in the control room or in a field equipment room. There are normally 2 communication links, and these may be used for communication with a host computer. They are referred to as Host 1 and Host 2.

The LTD communication links use RS485. To interface with the host computer using RS232, a converter must be used. The default baud rate is 9600 baud, and this cannot be changed at this time. The other parameters are odd parity, 8 data bits, and 1 stop bit. The transmission mode used is RTU.

The Modbus protocol requires the use of an address. The Tank ID in the LTD may be set using the touch screen user interface, and also serves as the Modbus address. If more than one LTD is connected on the same data bus, it is important to set a unique address for each unit.

4 Programming Suggestions

4.1 Introduction

To completely understand the operation of the LTD, the programmer should read the Operation Manual. While not all of the information is duplicated here, many relevant details are given to assist in the programming task. Below are some suggestions on how to use the information available from the LTD.

The primary uses of the LTD are 1) to obtain accurate liquid level information, and 2) to obtain temperature and density profile information that will enable the detection of unsafe layering conditions. Layers detected by the LTD are indicated by the Temperature and Density Deviation alarms.

The relative importance of the various status bits has been covered in the section above where the addresses of the status bits are given. Refer to that section for a better understanding of which indicators are most important for the operators to see.

4.2 Finding Liquid Level

After powering on the LTD, accurate liquid level information is available only after the probe has touched the bottom of the tank and been driven back to the liquid/vapor interface. When the system is powered off, the last known probe position and liquid level are stored in non-volatile memory, and the same values are recalled when the system is powered on, but the "Un-cal" indication appears, since it is possible that these values are no longer correct.

To clear the "Un-cal" indicator, it is necessary to touch the bottom of the tank. This will automatically set the probe position to the value that should be displayed at the bottom of the tank.

The LTD can be set to remain in Manual Mode or go directly into Cal Mode on power up. If it remains in Manual, a calibration should be initiated manually and after that it will remain in Auto Mode, measuring liquid level. With a system commissioned in a tank with liquid, that should be all that is necessary. The LTD will automatically go through the steps to find liquid level, switch to Auto mode and monitor the liquid level.

4.3 Profile Information

A profile is initiated by placing the controller into the Profile mode. This may be done via the touch screen user interface, the automatic interval timer, or a command sent from a host computer. Point 0 is always the bottom point, and the liquid level is the last point. The number of points collected is given in the "Number of Points" variable. The probe is stopped at each point for a specified wait time (called "dwell time") for stabilization. Temperature and density are measured, and then the probe is driven to the next point. The unit is returned to the AUTO mode when the profile is completed.

As a profile is being collected, the "Number of Points" variable is incremented after each new point. Using this variable makes it possible to import only valid data points. When the probe first touches the bottom of the tank at the beginning of a new profile all the previous data points in memory and the "Profile Complete" status indicator are cleared. When the profile is complete and all data is available, the "Profile Complete" status indicator is set.

4.4 Profile Graphing

The easiest way to see stratification of temperature and density in a tank of liquid is by graphing the data. In order for this graph to be meaningful, it must have a high resolution. The expected changes in the data are small, and the graph must be able to show those changes. It is recommended that the range of temperature and density be chosen carefully. One way to achieve the correct scaling is to make the values read at the first point at the bottom of the tank as the center of the graph, and then plot a deviation from those values as the level increases. On most LNG tanks, it should be suitable to show a deviation from the center of 5 kg/m³ or 0.3 lb/ft³.